KARTHIKEYAN VR

Software Engineer — Python / AI-ML / Full-Stack Systems

Chennai, Tamil Nadu, India | karthikeyanvr17@gmail.com | +91-7200082696 | linkedin.com/in/karthikeyan-vr | github.com/Karthikn-VR

PROFESSIONAL SUMMARY

Computer Science graduate (CGPA 8.6) with hands-on software development experience in Python, FastAPI, and Java, gained through a Software Developer Trainee role and applied engineering projects spanning backend systems, REST APIs, and AI/ML (computer vision, retrieval-augmented generation, embeddings). Comfortable across the full software development lifecycle — requirement analysis, coding, debugging, testing, and documentation — with a strong foundation in data structures, algorithms, and systems-level problem solving. Seeking to contribute to core software engineering functions in a technically rigorous, engineering-driven environment.

EDUCATION

B.E., Computer Science and Engineering — S.A. Engineering College, Chennai *2022 – 2026*

CGPA: 8.6

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, SQL
- **Backend & Systems:** FastAPI, REST API Design, JWT Authentication, OOP, Data Structures & Algorithms
- **AI / Machine Learning:** PyTorch, CNN, Computer Vision, FAISS, Sentence-Transformers, Embeddings, Retrieval-Augmented Generation (RAG)
- **Databases:** PostgreSQL, Supabase, SQL Querying
- **Frontend:** React.js, Next.js, HTML, CSS
- **Cloud & Tools:** AWS Cloud Foundations, AWS Cloud Operations, Git, Linux fundamentals, CI/CD Fundamentals
- **Practices:** SDLC, Debugging, Testing, Technical Documentation, Cross-functional Collaboration

PROFESSIONAL EXPERIENCE

Software Developer Trainee — Arunai Enterprises, Chennai, India *03/2026 – 05/2026*

- Contributed to NexServe, a full-stack food service management application, working across backend and frontend components within a structured SDLC.
- Developed and supported REST API endpoints in FastAPI (Python), covering core application workflows and business logic.
- Implemented JWT-based authentication and integrated PostgreSQL/Supabase for persistent, structured data storage.
- Debugged application issues across the stack and tested features end-to-end to verify functional correctness before release.
- Documented bugs, fixes, and testing outcomes, and supported deployment activities while collaborating closely with the engineering team.

PROJECTS

AI-Powered Knowledge Base & RAG System — Best Project Award | *Python, FastAPI, FAISS, Sentence-Transformers*

- Built a retrieval-augmented generation (RAG) pipeline covering document ingestion, text processing, and embedding generation.
- Implemented vector similarity search with FAISS to power a document retrieval system, and exposed it via FastAPI endpoints.
- Debugged and refined the retrieval pipeline to improve system reliability and relevance, earning the Best Project Award.

Advanced DeepVision — Skin Cancer Detection System | *Python, PyTorch, CNN, FastAPI, Next.js*

- Built an end-to-end system integrating a CNN-based image classification model with an inference API and frontend interface.
- Implemented image preprocessing and inference pipelines, with REST APIs exposing classification and confidence scoring.
- Added Grad-CAM explainability visualization and tested the full inference pipeline for reliable end-to-end integration.

NexServe — Full-Stack Food Service Management System | *React.js, FastAPI, Python, PostgreSQL/Supabase, REST APIs*

- Built a full-stack web application enabling customer and vendor workflows, including filtering, cart, and checkout logic.
- Designed REST APIs with FastAPI, integrated JWT authentication, and connected the app to a relational database backend.

CERTIFICATIONS

- CS50's Introduction to Artificial Intelligence with Python — Harvard CS50
- AWS Academy Graduate — Cloud Foundations and Cloud Operations
- NPTEL — Foundation of Cloud IoT Edge ML

ACHIEVEMENTS

- Best Project Award — AI-Powered Knowledge Base & RAG System
- 1st Prize — College Hackathon, for developing a full-stack web application
- 2nd Place — IEEE Day, Innovative Web Application